Practical LLMs & Agentic AI

Guest Lecture | University of Greenwich MSc Data Analytics & Management Mondweep Chakravorty & Bence Csernak Co-Founders, Agentics Foundation — London Chapter

Today's Journey

- Part 1 From ML to Agents (20 min)
- Part 2 Architecture of Agent Systems (20 min)
- Part 3 Live Demonstrations (30-40 min)
- Part 4 Governance & Responsible AI (20 min)
- Part 5 The Business Case & Future (15 min)
- Part 6 Q&A & Discussion (15 min)

About Us

Mondweep Chakravorty

Co-Founder, Agentics Foundation — London
Background: Group Lotus, Jaguar Land Rover
107 GitHub repos spanning agentic AI
CISO London Summit presenter
IC-ETSI 2025 — University of Greenwich

Bence Csernak

Co-Founder, Agentics Foundation — London
[Add Bence's bio details]

You Already Know the Building Blocks

Machine Learning basics — models learn from data
Prediction — models generate outputs from inputs
Agency — the theoretical concept of autonomous action

Today we bridge from theory to the real world:
What happens when you give a model tools, memory, and goals?

Part 1

From ML to Agents

The AI Assistance Spectrum

Level	Description	Example	Autonomy
Copilot	Suggests completions	Microsoft Copilot (your tool!)	Low
Assistant	Follows instructions	ChatGPT, Claude	Medium
Agent	Pursues goals autonomously	Claude Code, Forge agents	High
Swarm	Coordinated multi-agent teams	Claude-Flow, Agentic QE	Very High

What Makes an AI System an Agent?

Tools — agents can read files, call APIs, run tests, edit code
Memory — agents remember past outcomes and learn from them
Goals — agents pursue objectives, not just respond to prompts
Autonomy — agents decide their next action without step-by-step instruction
Coordination — agents can communicate and collaborate with other agents

Key distinction: reactive tools vs proactive agents that "accomplish goals, align stacked tasks, and complete them without direct supervision"

The PACT Framework — Classifying Agentic AI

The Four Dimensions

P — Proactive
Anticipate issues before they occur

A — Autonomous
Self-operating with safety mechanisms

C — Collaborative
Human expertise + AI capabilities

T — Targeted
Risk-based focus on high-impact areas

Think: SAE Levels for AI

Like autonomous vehicle levels (L0-L5), PACT classifies how autonomous an AI system is.

Your Copilot experience?
Low P, Low A, Medium C, Low T

A production agent swarm?
High P, High A, High C, High T

Your Copilot Is the Starting Point

Microsoft Copilot — you already use it under institutional licensing

Copilot today:
Single-agent assistance | Reactive suggestions | Human-driven

Where the industry is going:
Multi-agent orchestration | Proactive quality | Graduated autonomy

The same principles (tools, memory, goals) apply at every scale
Today we'll show you what's possible when you scale up

Part 2

Architecture of Agent Systems

How Agents Work — No Code Required

Perception — Agents read code, APIs, test results, user requirements
Reasoning — LLMs process context and decide next actions
Action — Agents execute through tools (file editing, API calls, tests)
Memory — Agents learn from outcomes and store patterns for reuse
Coordination — Agents communicate via shared memory or direct protocols

Think of it as a team of analysts:
each member has a speciality, shares a project dashboard,
and coordinates through stand-ups and handoffs.

Real Architecture: Claude-Flow

The leading open-source multi-agent orchestration platform

5-Layer Architecture:
1. Entry Layer — CLI and MCP servers with security hardening
2. Routing Layer — Q-Learning routers, 8 mixture-of-experts, 42+ skills
3. Swarm Coordination — hierarchical & peer-to-peer topologies
4. Agent Pool — 60+ specialized agents (coders, testers, reviewers...)
5. Intelligence Layer — self-optimization, vector search, 9 RL algorithms

Claude-Flow in Numbers

500K	60+	84.8%	2.8-4.4x
Downloads worldwide	Specialized Agents	SWE-Bench Solve Rate	Faster Task Completion

Agent Specialization — Forge

Forge: autonomous quality engineering with 8 specialized agents

Specification Verifier — checks requirements (Sonnet)
Test Runner — executes test suites (Haiku)
Failure Analyzer — diagnoses what went wrong (Sonnet)
Bug Fixer — writes the fix (Opus)
Quality Gate Enforcer — checks all gates pass (Haiku)
Accessibility Auditor — WCAG compliance (Sonnet)
Auto-Committer — commits passing code (Haiku)
Learning Optimizer — improves future runs (Sonnet)

Like a project team — each member matched to the right skill level

The Autonomous Quality Loop

Specify -> Test -> Analyze -> Fix -> Audit -> Gate -> Commit -> Learn

This loop runs continuously without human intervention:
Quality is forged in, not bolted on

But humans remain in the loop for:
Strategy and design decisions
Reviewing critical changes
Adjusting confidence thresholds
Overriding agent decisions when needed

Part 3

Live Demonstrations

Demo Lineup

#	Demo	Key Metric	Theme
D1	40-Minute App Build (Pre-Route)	5 agents, 40 min to working app	Speed to market
D2	AI Security Audit (MAESTRO)	23 vulns in 4 hrs, 16:1 ROI	Governance at speed
D3	AI Analytics (Auto-Analyst)	Real-time pricing intelligence	Your domain transformed
D4	Knowledge Graph (LBS)	3,963 nodes enriched for $14	AI at minimal cost
D5	WASM Image Filters (Live)	10-50x WASM vs JS speedup	Privacy by design
D6	Assam AI Governance (Live)	20-30 days to 5-7 days	Government transparency
D7	Driftwise (Live)	Voice-first, GPS-triggered	Voice-first AI

DEMO: 40-Minute App Build — Pre-Route

London Agentics Meetup | August 2025

[Screenshot placeholder]
github.com/mondweep/london-agentics-meetup-13aug-25

github.com/mondweep/london-agentics-meetup-13aug-25

5 specialized agents collaborated: Coordinator, Research, Backend, Frontend, Test Engineer

Built a complete traffic monitoring app for Kent residents in 40 minutes

Express.js + TypeScript backend with mock Google Maps/TomTom APIs

Business Takeaway: What would take weeks, done in under an hour — with quality built in

DEMO: AI Security Audit — MAESTRO Framework

CISO London Summit | Rela8 Group

[Screenshot placeholder]

github.com/mondweep/agentic-ai-security-demo-rela8group-ciso-london-summit

Analysed ElizaOS: 127 files, 42K+ lines
Found 23 vulnerabilities in 4 hours
(vs 15-20 days traditional audit)

3 Critical findings:
Plugin Supply Chain Attack (56/70)
scored highest risk

Cost: $5-15K vs $50-150K traditional

github.com/mondweep/agentic-ai-security-demo-rela8group-ciso-london-summit

Business Takeaway: Governance and security can be accelerated, not just development

Security Audit ROI

96%

Faster 4 hrs vs 15-20 days

90%

Cheaper $5-15K vs $50-150K

16:1

3-Year ROI mitigation value

23

Vulnerabilities found automatically

DEMO: AI-Powered Analytics — Auto-Analyst

AI transforming traditional data analytics workflows

[Screenshot placeholder]
github.com/mondweep/Auto-Analyst

github.com/mondweep/Auto-Analyst

Automotive dealership analytics platform with AI-powered insights

Interactive dashboards
Pricing intelligence
Market opportunity analysis

This is YOUR domain —
data analytics transformed by agents

Business Takeaway: AI transforms traditional analytical workflows — and data analysts are best positioned to lead

DEMO: University Knowledge Graph — LBS

AI-driven content intelligence at minimal cost

[Screenshot placeholder]
github.com/mondweep/university-pitch

github.com/mondweep/university-pitch

Semantic enrichment of london.edu
3,963 nodes, 3,953 edges

Phases: Crawling -> Domain Modeling -> Semantic Enrichment

Sentiment analysis, topic extraction, persona classification

Total cost: ~$14

Business Takeaway: AI applied to a university — like Greenwich. Near-zero marginal cost for content intelligence

DEMO: WASM Image Filters

Privacy-first processing — your data never leaves the browser

[Screenshot placeholder]
https://wasm-tinkering.netlify.app/

https://wasm-tinkering.netlify.app/

- Upload an image, apply 10+ filters
  JS vs WASM benchmark side-by-side
- 10-50x performance improvement
  with Rust/WASM over JavaScript
- ALL processing runs locally —
  data never leaves the device
- Built with agentic AI tools

Business Takeaway: Privacy-by-design and performance optimization are not mutually exclusive

DEMO: LuitPlayer — Grand Piano

7-agent swarm built a 7-octave piano with sheet music OMR

[Screenshot placeholder]
https://grand-piano-thisismon.netlify.app/

https://grand-piano-thisismon.netlify.app/

88 keys (C1-B7), 5 timbres
Built by 7-agent Claude Flow swarm:
Coordinator, Frontend, Audio,
OMR, Test, Deploy, Integration

Optical Music Recognition for
Assamese sheet music (Tesseract WASM)

Multi-touch, sustain pedal, mixer

Business Takeaway: Multi-agent collaboration can produce sophisticated creative applications

DEMO: Kumno — Khasi Travel Companion

AI translation enabling cultural preservation and tourism

[Screenshot placeholder]
https://kumno.netlify.app/

https://kumno.netlify.app/

80+ Khasi phrases with phonetic guides
8 categories: Greetings, Taxis, Markets, Food, Directions...

Bah/Kong honorific toggle (cultural sensitivity built in)

AI-powered real-time translation
Fully offline-capable PWA

Business Takeaway: AI translation bridging language barriers while preserving cultural nuance

DEMO: Assam AI Governance

AI for government transparency and anti-fraud

[Screenshot placeholder]
https://assam-ai-usecase-governance-pwa.netlify.app/

https://assam-ai-usecase-governance-pwa.netlify.app/

Property Registration Digitalization:
20-30 days reduced to 5-7 days
80% remote completion target

Infrastructure Cost Auditing:
AI anomaly detection (Isolation Forest)
Catches 3x cost inflation
50+ crore INR annual savings target

Trilingual: English, Assamese, Hindi

Business Takeaway: AI applied directly to government governance and transparency — measurable cost savings

DEMO: Driftwise — Discover Interesting Facts

Voice-first, driver-safe AI that transforms commutes into discovery

[Screenshot placeholder]
https://driftwise-discover-interesting-facts.netlify.app/

https://driftwise-discover-interesting-facts.netlify.app/

GPS-triggered historical fact delivery
No user interaction required

Voice-first design — no screen glancing
5-second voice command window

Anti-generic filtering: rejects clichés, surfaces named individuals, specific dates, unusual events

Gemini Live API for real-time voice

Business Takeaway: Voice-first design pattern — responsible AI with driver safety as the primary constraint

Part 4

Governance & Responsible AI

Let's Be Honest — Current Limitations

Weak business-rule understanding
Agents can miss domain-specific nuances that humans catch intuitively

Model reliability issues
LLMs can hallucinate, lose context, or produce inconsistent outputs

Over-engineering tendencies
Agents sometimes add unnecessary complexity

Severity calibration challenges
Distinguishing critical issues from minor ones remains difficult

Real maintenance overhead
Agent systems require monitoring, tuning, and governance

"These are engineering trade-offs, not showstoppers. Critical thinking — not hype — is what governance requires."

— The honest assessment approach

MAESTRO — 7-Layer Security Framework for Agentic AI

Layer	Focus	Example Concerns
1. Foundational Models	LLM integrations, inference	Prompt security, model poisoning
2. Data Operations	Databases, RAG, vectors	Data leakage, poisoned embeddings
3. Agent Frameworks	Orchestration, decision logic	Rogue agents, decision manipulation
4. Deployment & Infra	Runtime, APIs, networking	Unauthorized access, API abuse
5. Evaluations	Logging, monitoring, testing	Blind spots, inadequate audit trails
6. Security & Compliance	Auth, secrets, policies	Credential theft, policy violations
7. Agent Ecosystem	Plugins, actions, tools	Supply chain attacks, malicious plugins

Quality Gates — Guardrails for Autonomous Agents

7 blocking gates from Forge that agents must pass:

1. Functional Correctness — code compiles and passes unit tests
2. Behavioural Compliance — Gherkin specifications all pass
3. Code Coverage — meets minimum thresholds (85% baseline, 95% critical)
4. Security Violations — no SAST/DAST findings above threshold
5. API Contract Validation — responses match expected contracts
6. Accessibility — WCAG AA compliance
7. Resilience — chaos testing passes

Agents operate freely within these boundaries
Gates are the governance mechanism, not humans watching every step

Confidence Tiers — Trust Earned Through Evidence

How do you know when to trust an agent with more autonomy?

Platinum (10+ successes) — fully autonomous operation
Gold (3-9 successes) — autonomous with monitoring
Silver (1-2 successes) — requires human review

Trust is earned incrementally through empirical evidence
Not assigned by role or assumption

Like vendor performance ratings in procurement —
you wouldn't give a new vendor a £1M contract on day one

TinyDancer — Cost Governance Through Model Routing

Complexity	Model	Use Cases	Cost
0-20 (Simple)	Haiku	Syntax fixes, type additions	Lowest
20-70 (Standard)	Sonnet	Bug fixes, test generation	Moderate
70-100 (Complex)	Opus	Architecture, security analysis	Highest

Cost Governance Impact

75%

Token Cost Reduction

3

Model Tiers Intelligent routing

85%

Multi-Provider Cost Savings

Human-in-the-Loop — A Design Principle, Not a Limitation

Agents augment, never fully replace human judgement

Humans own:
Strategy & design decisions
Critical business-rule validation
Ethical oversight & escalation
Adjusting confidence thresholds

Agents own:
Execution at speed and scale
Pattern recognition across large codebases
Continuous quality monitoring
Routine decision-making within guardrails

Discussion

"If you were deploying an agentic AI system in your organisation, what quality gates would you define?"

"How would you measure whether an agent has earned enough trust for increased autonomy?"

Part 5

The Business Case & Future

ROI Evidence — Numbers You Can Take to a Board Meeting

Application	Traditional	AI-Driven	Improvement
Security Audit	15-20 days, $50-150K	4 hours, $5-15K	96% faster, 90% cheaper
Task Completion	Baseline	2.8-4.4x faster	Up to 340% improvement
Token Costs	Baseline	75% reduction	Via model routing
Content Enrichment	Manual	~$14 for 3,963 nodes	Near-zero marginal cost
App Development	Weeks/months	40 minutes (Pre-Route)	Radical speed to market
Test Coverage	Manual QA	85-95% automated	Quality gates enforce standards

Cross-Industry Applications

Already Happening

Healthcare — diagnostic AI, care coordination
Automotive — pricing intelligence, inventory
Education — recruitment, Socratic tutors
Public Sector — fraud prevention, governance
Security — automated vulnerability assessment
Media — content discovery, sentiment analysis

Where It's Going

Single agents -> coordinated swarms
Reactive testing -> proactive quality
Manual governance -> automated compliance
Human-only decisions -> graduated autonomy
Agent-Ready Web (ARW) — websites designed for agent consumption
A2A protocols — agents from different vendors collaborating

Agentic AI Is Already Transforming Universities

Recruitment — 24/7 digital concierges handling credit evaluations and campus tours
Academic Support — Socratic AI tutors generating practice problems in real time
Predictive Intervention — identifying struggling students before midterms
Mental Health — triage agents offering coping strategies, escalating serious cases
Finance — automated invoice processing, grant identification, proposal drafting
Compliance — regulatory reporting generating audit-ready documents

"Institutions operationalising AI early will widen their performance gap"
— Inside Higher Ed, January 2026

Your Role as Data Analysts

You are uniquely positioned for this revolution

Your superpowers:
Metrics & measurement — you know how to evaluate effectiveness
Critical analysis — you can spot when AI claims don't hold up
Business context — you bridge technical capability and business value
Governance thinking — data governance skills transfer directly

The governance challenge needs business-minded people,
not just engineers

Data analysts will evaluate, govern, and propose agentic AI systems

Part 6

Q&A & Discussion

Discussion Topics

1. What business processes in your industry could benefit from agentic AI?
2. How would you design governance for an agent that manages financial data?
3. What ethical considerations arise when agents learn from outcomes?
4. How does your experience with Microsoft Copilot map to the PACT framework?
5. If you could deploy one agent in your workplace tomorrow, what would it do?

Glossary — Quick Reference

Term	Business-Friendly Definition
Agent	AI system that pursues goals autonomously — a digital team member
Swarm	Coordinated team of AI agents — a project team with specialized roles
Quality Gate	Checkpoint that work must pass — like approval stages in a process
Confidence Tier	Trust level based on track record — like vendor performance ratings
Model Routing	Directing tasks to right-sized AI — assigning work by complexity
PACT	Framework for classifying agent autonomy: Proactive, Autonomous, Collaborative, Targeted
MAESTRO	7-layer security framework for evaluating agentic AI systems
MCP	Model Context Protocol — standard way AI agents connect with tools (USB for AI)

Resources & Further Reading

Open-Source Repos

Claude-Flow — github.com/ruvnet/claude-flow
Forge — github.com/ikennaokpala/forge
Agentic QE — github.com/proffesor-for-testing/agentic-qe
Pre-Route — github.com/mondweep/london-agentics-meetup-13aug-25
MAESTRO Demo — github.com/mondweep/agentic-ai-security-demo...
Auto-Analyst — github.com/mondweep/Auto-Analyst
University KG — github.com/mondweep/university-pitch

Live Demo Apps

WASM Filters — wasm-tinkering.netlify.app
LuitPlayer — grand-piano-thisismon.netlify.app
Kumno — kumno.netlify.app
Assam Gov — assam-ai-usecase-governance-pwa.netlify.app
Driftwise — driftwise-discover-interesting-facts.netlify.app

Frameworks

PACT — forge-quality.dev
Agentic QE — agentic-qe.dev

Agentics Foundation — London Chapter

- Co-founded by Mondweep Chakravorty and Bence Csernak
- Global chapters: London, Serbia, India (Assam), New Zealand
- Partnership with University of Greenwich — IC-ETSI 2025 conference
- Regular meetups with hands-on workshops on Agentic Engineering Practices
- Hackathons proving the viability of agent-assisted rapid development
- Get involved: join our London meetups and hackathons

Thank You

Mondweep Chakravorty & Bence Csernak Co-Founders, Agentics Foundation — London Chapter github.com/mondweep Questions?